

Data Fusion with Covariates

Roshni Sahoo
r.sahoo@columbia.edu

Columbia University

1 Statistical Setup

Let S_t, P_t denote the survey and target population distributions respectively at time-step t . Let P_t be a distribution over a covariate $X \in \mathcal{X}$, outcome $Y \in \mathcal{Y}$, and a binary response indicator $R \in \{0, 1\}$. Let S_t be the $X, Y \mid R = 1$ marginal of P_t . We have access to S_t for $t = 1, \dots, M$ and P_t for $t = 1, \dots, m$, where $m < M$. For simplicity, we will assume that the covariate distribution of the survey and population distributions are the same.

We aim to learn a decision rule h that performs well under the target population P_t at the current time-step $t = M$.

$$\min_{h: \mathcal{X} \rightarrow \mathbb{R}} \mathbb{E}_{P_t} [L(h(X), Y)].$$

We define relative probability of sample selection between two individuals with outcome y and y' at time-step t as

$$w(x, y, y'; t) = \frac{\mathbb{P}_{P_t} [R = 1 \mid X = x, Y = y]}{\mathbb{P}_{P_t} [R = 1 \mid X = x, Y = y']}. \quad (1)$$

Assumption 1 (Stationary Selection). The function $w(x, y, y'; t)$ is constant in t .

Lemma 1. *The population and survey conditional distributions are related via a time-invariant exponential tilting, i.e. there exists some $\theta : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ such that*

$$dP_{t,Y|X}(y) = dS_{t,Y|X}(y) \cdot \exp(\theta(x, y) - A_{t,x}(\theta)),$$

where $A_{t,x}(\theta) = \log \left(\int \exp(\theta(x, y)) \cdot dS_{t,Y|X=x}(y) dy \right)$, if and only if Assumption 1 holds.

This result implies that

$$\mathbb{E}_{P_t} [L(h(X), Y)] = \mathbb{E}_{S_{t,X}} \left[\mathbb{E}_{S_{t,Y|X}} [\exp(\theta(X, Y)) \cdot L(h(X), Y) \mid X] \middle/ \mathbb{E}_{S_{t,Y|X}} [\exp(\theta(X, Y)) \mid X] \right].$$

2 Algorithm

The time-invariant function that guarantees unbiasedness under stationarity can be learned by leveraging the data from the paired survey and target distributions S_t, P_t when $t \in [m]$.

Define a pooled distribution F over paired data from S_t, P_t for $t = 1, 2, \dots, m$. Let T be a random variable that denotes the time step and let Z be a pseudolabel that captures whether a sample is drawn from the survey or target population. The distribution F is a distribution over (X, Y, Z, T) , where $Z \sim \text{Bernoulli}(1/2)$ and $T \sim \text{Uniform}([m])$, $F_{X,Y|Z=1,T=t} = P_t$, and

$F_{X,Y|Z=0,T=t} = S_t$. Let Θ be the space of square-integrable functions with measurable functions with respect to F_Y , i.e. $\Theta := L^2(F_{X,Y}, \mathcal{X} \times \mathcal{Y})$. Let $\mathcal{A} \subset L^2(F_{X,T}, \mathcal{X} \times \mathbb{R})$ with last coordinate equal to zero, i.e. $\mathcal{A} = \{\alpha \in \mathbb{R}^m \mid \alpha(\cdot, m) = 0\}$. Define a loss function

$$L(\theta, a, z) := (1 - z) \cdot \exp(\theta + a) - z \cdot (\theta + a). \quad (2)$$

Our goal is to obtain θ from Lemma 1 such that $\mathbb{E}_{S_M} [\theta(X, Y)] = 1$
Implementation

1. Enforce the normalization constraint post-hoc. First, solve

$$\{\tilde{\theta}(\cdot), \tilde{\alpha}(\cdot)\} \in \arg \min_{(\theta, \alpha) \in \Theta \times \mathcal{A}} \{\mathbb{E}_F [L(\theta(X, Y), \alpha(X, T), Z)] : \alpha(\cdot; m) = 0\}.$$

Note that the normalization constraint does not depend on S_M . Second, given samples $(X, Y) \sim S_M$ and define $\tilde{Y} = \exp(\theta(X, Y))$.

$$c \in \arg \min_{c: \mathcal{X} \rightarrow \mathbb{R}} \mathbb{E}_{S_M} [(\tilde{Y} - c(X))^2].$$

Third, define

$$\theta(x, y) := \tilde{\theta}(x, y) - c(x).$$

$$\mathbb{E}_{P_t} [L(h(X), Y)] = \mathbb{E}_{S_t} [\exp(\tilde{\theta}(X, Y)) \cdot L(h(X), Y)]. \quad (3)$$